1. Name of the medicinal product
EZESIM 10/10 (Ezetimibe and Simvastatin Tablets)

2. Qualitative and quantitative composition
Each Tablet contains : Ezetimibe USP 10mg
Simvastatin Ph.Eur 10 mg

3. Pharmaceutical form
Tablet.
White to off white, capsule shaped tablets debossed with "A029" on one side and plain on other side.

4. Clinical particulars
4.1 Therapeutic indications
Prevention of Cardiovascular Events
EZESIM is indicated to reduce the risk of cardiovascular events (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or need for revascularization), in patients with coronary heart disease (CHD) and history of acute coronary syndrome (ACS), either previously treated with a statin or not.

Primary Hypercholesterolemia
EZESIM is indicated as adjunctive therapy to diet for the reduction of elevated total cholesterol (total-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), triglycerides (TG), and non-high-density lipoprotein cholesterol (non-HDL-C), and to increase high-density lipoprotein cholesterol (HDL-C) in patients with primary (heterozygous familial and non-familial) hypercholesterolemia or mixed hyperlipidemia.

Homozygous Familial Hypercholesterolemia (HoFH)
EZESIM is indicated for the reduction of elevated total-C and LDL-C levels in patients with HoFH. Patients may also receive adjunctive treatments (e.g., LDL apheresis).

4.2 Posology and method of administration
The patient should be placed on a standard cholesterol-lowering diet before receiving EZESIM and should continue on this diet during treatment with EZESIM. The dosage should be individualized according to the baseline LDL-C level, the recommended goal of therapy, and the patient's response. EZESIM should be taken as a single daily dose in the evening, with or without food.

Patients with Primary Hypercholesterolemia
In patients with primary hyperlipidemia or mixed hyperlipidemia, the dosage range is 10/10 mg/day through 10/80 mg/day. The recommended usual starting dose is 10/20 mg/day. Initiation of therapy with 10/10 mg/day may be considered for patients requiring less aggressive LDL-C reductions. Patients who require a larger reduction in LDL-C (greater than 55%) may be started at 10/40 mg/day. After initiation or titration of EZESIM, lipid levels may be analyzed after 2 or more weeks and dosage adjusted, if needed. The 10/80 mg dose of EZESIM is only recommended in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks (see section 4.4. Special warnings and precautions for use, Myopathy/Rhabdomyolysis).

For patients who are currently on both simvastatin and ezetimibe, no dosage adjustment is required when switching to EZESIM.

Patients with Coronary Heart Disease
In the cardiovascular events risk reduction study (IMPROVE-IT), the starting dose was 10/40 mg once a day in the evening. The 10/80-mg dose is only recommended when the benefits are expected to outweigh the potential risks. (See section 4.4. Special warnings and precautions for use.)

Patients with Homozygous Familial Hypercholesterolemia
The recommended dosage for patients with homozygous familial hypercholesterolemia is EZESIM 10/40 mg/day or 10/80 mg/day in the evening. The 10/80-mg dose is only recommended when the benefits are expected to outweigh the potential risks (see below section 4.3. Contraindications, and section 4.4. Special warnings and precautions for use, Myopathy/Rhabdomyolysis). EZESIM should be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients or if such treatments are unavailable.

In patients taking lomitapide concomitantly with EZESIM, the dose of EZESIM should not exceed 10/40 mg/day (see sections 4.4. Special warnings and precautions for use, Myopathy/Rhabdomyolysis and 4.5 Interaction with other medicinal products and other forms of interaction).

Use in the Elderly
No dosage adjustment is required for elderly patients (see section 5.2, Special Populations).

Use in Pediatric Patients
Treatment with EZESIM is not recommended.

Use in Hepatic Impairment
No dosage adjustment is required in patients with mild hepatic insufficiency (Child-Pugh score 5 or 6). Treatment with EZESIM is not recommended in patients with moderate (Child-Pugh score 7 to 9) or severe (Child-Pugh score > 9) liver dysfunction (see sections 4.4. Special warnings and precautions for use. and 5.2, Special Populations).

Use in Renal Impairment
No dosage adjustment is required for patients with moderate renal insufficiency. If treatment in patients with severe renal insufficiency (creatinine clearance ≤ 30 mL/min) is deemed necessary, dosages above 10/10 mg/day should be implemented cautiously (see section 5.2, Special Populations).

Coadministration with other medicines
Dosing of EZESIM should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.
In patients taking amiodarone, verapamil, diltiazem, or products containing elbasvir or grazoprevir concomitantly with EZESIM, the dose of EZESIM should not exceed 10/20 mg/day (see 4.4. Special warnings and precautions for use, Myopathy/Rhabdomyolysis and 4.5 Interaction with other medicinal products and other forms of interaction).

In patients taking amlodipine concomitantly with EZESIM, the dose of EZESIM should not exceed 10/40 mg/day (see sections 4.4. Special warnings and precautions for use, Myopathy/Rhabdomyolysis and 4.5 Interaction with other medicinal products and other forms of interaction).

The safety and effectiveness of EZESIM administered with fibrates have not been studied. Therefore, the combination of EZESIM and fibrates should be avoided (see 4.3. Contraindications, 4.4. Special warnings and precautions for use, Myopathy/Rhabdomyolysis and 4.5 Interaction with other medicinal products and other forms of interaction).

Ezetimibe/Simvastatin is available in strengths of 10/10, 10/40, 10/80. However, only Ezesim 10mg/10mg is marketed in Malaysia. Therefore, brand Ezesim is not able to deliver all approved dose regimes of Ezetimib/Simvastatin

4.3 Contraindications
-Hypersensitivity to the active substance(s) or to any of the excipients
-Pregnancy and lactation (see section 4.6).
-Active liver disease or unexplained persistent elevations in serum transaminases.

-Concomitant administration of potent CYP3A4 inhibitors (agents that increase AUC) (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, nefazodone, and drugs containing cobicistat) (see sections 4.4 and 4.5).
-Concomitant administration of gemfibrozil, ciclosporin, or danazol (see sections 4.4 and 4.5).
In patients with HoFH, concomitant administration of lomitapide with doses > 10/40 mg EZESIM (see sections 4.2, 4.4 and 4.5).

4.4 Special warnings and precautions for use

Myopathy/Rhabdomyolysis

In post-marketing experience with ezetimibe, cases of myopathy and rhabdomyolysis have been reported. Most patients who developed rhabdomyolysis were taking a statin concomitantly with ezetimibe. However, rhabdomyolysis has been reported very rarely with ezetimibe monotherapy and very rarely with the addition of ezetimibe to other agents known to be associated with increased risk of rhabdomyolysis.
EZESIM contains simvastatin. Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness or weakness with creatine kinase (CK) above 10 X the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and very rare fatalities have occurred. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma (i.e., elevated simvastatin and simvastatin acid plasma levels), which may be due, in part, to interacting drugs that interfere with simvastatin metabolism and/or transporter pathways (see section 4.5).
As with other HMG-CoA reductase inhibitors, the risk of myopathy/rhabdomyolysis is dose related for simvastatin.
The risk of myopathy is greater in patients on EZESIM 10/80 mg compared with other statin-based therapies with similar LDL-C-lowering efficacy. Therefore, the 10/80-mg dose of EZESIM should only be used in patients with severe hypercholesterolaemia and at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. In patients taking EZESIM 10/80 mg for whom an interacting agent is needed, a lower dose of EZESIM or an alternative statin-based regimen with less potential for drug-drug interactions should be used (see below *Measures to reduce the risk of myopathy caused by medicinal product interactions* and sections 4.2, 4.3, and 4.5).
Caution should be used when prescribing EZESIM to Asian patients and the lowest dose necessary should be employed.

Reduced function of transport proteins

Reduced function of hepatic OATP transport proteins can increase the systemic exposure of simvastatin acid and increase the risk of myopathy and rhabdomyolysis. Reduced function can occur as the result of inhibition by interacting medicines (eg ciclosporin) or in patients who are carriers of the SLCO1B1 c.521T>C genotype.
Patients carrying the SLCO1B1 gene allele (c.521T>C) coding for a less active OATP1B1 protein have an increased systemic exposure of simvastatin acid and increased risk of myopathy. The risk of high dose (80 mg) simvastatin related myopathy is about 1 % in general, without genetic testing. Homozygote C allele carriers (also called CC) treated with 80 mg have a risk of myopathy within one year, while the risk in heterozygote C allele carriers (CT) is 1.5%. The corresponding risk is 0.3% in patients having the most common genotype (TT) (see section 5.2). Where available, genotyping for the presence of the C allele should be considered as part of the benefit-risk assessment prior to prescribing 80 mg simvastatin for individual patients and high doses avoided in those found to carry the CC genotype. However, absence of this gene upon genotyping does not exclude that myopathy can still occur.

Creatine Kinase measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of any plausible alternative cause of CK increase as this makes value interpretation difficult. If CK levels are significantly elevated at baseline (>5 X ULN), levels should be re-measured within 5 to 7 days later to confirm the results.

Before the treatment

All patients starting therapy with EZESIM, or whose dose of EZESIM is being increased, should be advised of the risk of myopathy and told to report promptly any unexplained muscle pain, tenderness or weakness.

Caution should be exercised in patients with pre-disposing factors for rhabdomyolysis. In order to establish a reference baseline value, a CK level should be measured before starting treatment in the following situations:

• Elderly (age ≥65 years)
• Female gender
• Renal impairment
• Uncontrolled hypothyroidism
• Personal or familial history of hereditary muscular disorders
• Previous history of muscular toxicity with a statin or fibrate
• Alcohol abuse.

In such situations, the risk of treatment should be considered in relation to possible benefit, and clinical monitoring is recommended. If a patient has previously experienced a muscle disorder on a fibrate or a statin, treatment with any statin-containing product (such as EZESIM) should only be initiated with caution. If CK levels are significantly elevated at baseline (>5 X ULN), treatment should not be started.

Whilst on treatment

If muscle pain, weakness or cramps occur whilst a patient is receiving treatment with EZESIM, their CK levels should be measured. If these levels are found, in the absence of strenuous exercise, to be significantly elevated (>5 X ULN), treatment should be stopped. If muscular symptoms are severe and cause daily discomfort, even if CK levels are <5 X ULN, treatment discontinuation may be considered. If myopathy is suspected for any other reason, treatment should be discontinued.

There have been very rare reports of an immune-mediated necrotising myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterised by persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment (see section 4.8).

If symptoms resolve and CK levels return to normal, then re-introduction of EZESIM or introduction of another statin-containing product may be considered at the lowest dose and with close monitoring.

A higher rate of myopathy has been observed in patients titrated to the 80 mg dose of simvastatin (see section 5.1). Periodic CK measurements are recommended as they may be useful to identify subclinical cases of myopathy. However, there is no assurance that such monitoring will prevent myopathy.

Therapy with EZESIM should be temporarily stopped a few days prior to elective major surgery and when any major medical or surgical condition supervenes.

Measures to reduce the risk of myopathy caused by medicinal product interactions (see also section 4.5)

The risk of myopathy and rhabdomyolysis is significantly increased by concomitant use of EZESIM with potent inhibitors of CYP3A4 (such as itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, nefazodone, and medicinal products containing cobicistat), as well as ciclosporin, danazol, and gemfibrozil. Use of these medicinal products is contraindicated (see section 4.3).

Due to the simvastatin component of EZESIM, the risk of myopathy and rhabdomyolysis is also increased by concomitant use of other fibrates, lipid-lowering doses (≥1 g/day) of niacin or by concomitant use of amiodarone, amlodipine, verapamil or diltiazem with certain doses of EZESIM (see sections 4.2 and 4.5). The risk of myopathy including rhabdomyolysis may be increased by

concomitant administration of fusidic acid with EZESIM. For patients with HoFH, this risk may be increased by concomitant use of lomitapide with EZESIM (see section 4.5).
Consequently, regarding CYP3A4 inhibitors, the use of EZESIM concomitantly with itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin, nefazodone, and medicinal products containing cobicistat is contraindicated (see sections 4.3 and 4.5). If treatment with potent CYP3A4 inhibitors (agents that increase AUC) is unavoidable, therapy with EZESIM must be suspended (and use of an alternative statin considered) during the course of treatment. Moreover, caution should be exercised when combining EZESIM with certain other less potent CYP3A4 inhibitors: fluconazole, verapamil, diltiazem (see sections 4.2 and 4.5). Concomitant intake of grapefruit juice and EZESIM should be avoided.
Simvastatin must not be co-administered with systemic formulations of fusidic acid or within 7 days of stopping fusidic acid treatment. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving fusidic acid and statins in combination (see section 4.5). The patient should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness.
Statin therapy may be re-introduced seven days after the last dose of fusidic acid. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g. for the treatment of severe infections, the need for co-administration of EZESIM and fusidic acid should only be considered on a case-by-case basis under close medical supervision.
The combined use of EZESIM at doses higher than 10/20 mg daily with lipid-lowering doses (≥ 1 g/day) of niacin should be avoided unless the clinical benefit is likely to outweigh the increased risk of myopathy (see sections 4.2 and 4.5).
Rare cases of myopathy/rhabdomyolysis have been associated with concomitant administration of HMG-CoA reductase inhibitors and lipid-modifying doses (≥ 1 g/day) of niacin (nicotinic acid), either of which can cause myopathy when given alone.
In patients at high risk of cardiovascular disease and with well-controlled LDL-C levels on simvastatin 40 mg/day with or without ezetimibe 10 mg, there was no incremental benefit on cardiovascular outcomes with the addition of lipid-modifying doses (≥ 1 g/day) of niacin (nicotinic acid). Therefore, physicians contemplating combined therapy with simvastatin and lipid-modifying doses (≥ 1 g/day) of niacin (nicotinic acid) or products containing niacin should carefully weigh the potential benefits and risks and should carefully monitor patients for any signs and symptoms of muscle pain, tenderness, or weakness, particularly during the initial months of therapy and when the dose of either medicinal product is increased.
In addition, because the incidence of myopathy is higher in Chinese than in non-Chinese patients, co-administration of EZESIM with lipid-modifying doses (≥ 1 g/day) of niacin (nicotinic acid) is not recommended in Asian patients.
Acipimox is structurally related to niacin. Although there is no data on acipimox, the risk for muscle related toxic effects may be similar to niacin.
The combined use of EZESIM at doses higher than 10/20 mg daily with amiodarone, amlodipine, verapamil, or diltiazem should be avoided. In patients with HoFH, the combined use of EZESIM at doses higher than 10/40 mg daily with lomitapide must be avoided (see sections 4.2, 4.3 and 4.5).
Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 at therapeutic doses concomitantly with EZESIM, particularly higher EZESIM doses, may have an increased risk of myopathy. When co-administering EZESIM with a moderate inhibitor of CYP3A4 (agents that increase AUC), a dose adjustment may be necessary. For certain moderate CYP3A4 inhibitors, e.g. diltiazem, a maximum dose of 10/20 mg EZESIM is recommended (see section 4.2).
Simvastatin is a substrate of the Breast Cancer Resistant Protein (BCRP) efflux transporter. Concomitant administration of products that are inhibitors of BCRP (e.g. elbasvir and grazoprevir) may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy; therefore a dose adjustment of simvastatin should be considered depending on the prescribed dose. There is no data on Co-administration of elbasvir and grazoprevir with simvastatin; however, **the dose of EZESIM should not exceed 10/20 mg daily in patients receiving concomitant medication with products containing elbasvir or grazoprevir** (see section 4.5).

The is no safety and efficacy data of EZESIM administered with fibrates. There is an increased risk of myopathy when simvastatin is used concomitantly with fibrates (especially gemfibrozil). Therefore, concomitant use of EZESIM with gemfibrozil is contraindicated (see section 4.3) and concomitant use with other fibrates is not recommended (see section 4.5).

Daptomycin
Cases of myopathy and/or rhabdomyolysis have been reported with HMG-CoA reductase inhibitors (e.g. simvastatin and ezetimibe/simvastatin) co-administered with daptomycin. Caution should be used when prescribing HMG-CoA reductase inhibitors with daptomycin, as either agent can cause myopathy and/or rhabdomyolysis when given alone. Consideration should be given to temporarily suspend EZESIM in patients taking daptomycin unless the benefits of concomitant administration outweigh the risk. Consult the prescribing information of daptomycin to obtain further information about this potential interaction with HMG-CoA reductase inhibitors (e.g. simvastatin and ezetimibe/simvastatin) and for further guidance related to monitoring. (See section 4.5).

Liver Enzymes
In patients receiving ezetimibe with simvastatin, consecutive transaminase elevations (≥ 3 X ULN) have been observed (see section 4.8).
It is recommended that liver function tests be performed before treatment with EZESIM begins and thereafter when clinically indicated. Patients titrated to the 10/80-mg dose should receive an additional test prior to titration, 3 months after titration to the 10/80-mg dose, and periodically thereafter (e.g. semiannually) for the first year of treatment. Special attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise to 3 X ULN and are persistent, the drug should be discontinued. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy (see above *Myopathy/Rhabdomyolysis)*.
There have been rare post-marketing reports of fatal and non-fatal hepatic failure in patients taking statins, including simvastatin. If serious liver injury with clinical symptoms and/or hyperbilirubinaemia or jaundice occurs during treatment with EZESIM promptly interrupt therapy. If an alternate aetiology is not found, do not restart EZESIM.
EZESIM should be used with caution in patients who consume substantial quantities of alcohol.

Hepatic impairment
Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic impairment, EZESIM is not recommended (see section 5.2).

Diabetes mellitus
Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI > 30 kg/m^2, raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

Paediatric population
There was generally no detectable effect on growth or sexual maturation in the adolescent boys or girls (patients 10 to 17 years of age with heterozygous familial hypercholesterolaemia), or any effect on menstrual cycle length in girls. However, the is no data for effects of ezetimibe for a treatment period > 33 weeks on growth and sexual maturation (see sections 4.2 and 4.8).
The is no safety and efficacy data of ezetimibe co-administered with doses of simvastatin above 40mg daily in paediatric patients 10 to 17 years of age.
There is no data of use of Ezetimibe in patients younger than 10 years of age or in pre-menarchal girls (see sections 4.2 and 4.8).

The is no long-term efficacy of therapy data with ezetimibe in patients below 17 years of age to reduce morbidity and mortality in adulthood.

Fibrates
The safety and efficacy of ezetimibe administered with fibrates have not been established (see above and sections 4.3 and 4.5).
Anticoagulants
If EZESIM is added to warfarin, another coumarin anticoagulant, or fluindione, the International Normalised Ratio (INR) should be appropriately monitored (see section 4.5).

Interstitial lung disease
Cases of interstitial lung disease have been reported with some statins, including simvastatin, especially with long term therapy (see section 4.8). Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, EZESIM therapy should be discontinued.

Excipients
Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.
Ezesim contains less than 1 mmol (23 mg) sodium per tablet, that is to say essentially sodium-free.

4.5 Interaction with other medicinal products and other forms of interaction
Multiple mechanisms may contribute to potential interactions with HMG Co-A reductase inhibitors. Drugs or herbal products that inhibit certain enzymes (e.g. CYP3A4) and/or transporter (e.g. OATP1B) pathways may increase simvastatin and simvastatin acid plasma concentrations and may lead to an increased risk of myopathy/rhabdomyolysis.
Consult the prescribing information of all concomitantly used drugs to obtain further information about their potential interactions with simvastatin and/or the potential for enzyme or transporter alterations and possible adjustments to dose and regimens.

Pharmacodynamic interactions
Interactions with lipid-lowering medicinal products that can cause myopathy when given alone
The risk of myopathy, including rhabdomyolysis, is increased during concomitant administration of simvastatin with fibrates. Additionally, there is a pharmacokinetic interaction of simvastatin with gemfibrozil resulting in increased simvastatin plasma levels (see below *Pharmacokinetic interactions* and sections 4.3 and 4.4). Rare cases of myopathy/rhabdomyolysis have been associated with simvastatin co-administered with lipid-modifying doses (≥ 1 g/day) of niacin (see section 4.4).
Fibrates may increase cholesterol excretion into the bile, leading to cholelithiasis. In a preclinical study in dogs, ezetimibe increased cholesterol in the gallbladder bile. Although the relevance of this preclinical finding to humans is unknown, co-administration of EZESIM with fibrates is not recommended (see section 4.4).

Pharmacokinetic interactions
Prescribing recommendations for interacting agents are summarised in the table below (further details are provided in the text; see also sections 4.2, 4.3, and 4.4).

Drug Interactions Associated with Increased Risk of Myopathy/Rhabdomyolysis	
Interacting agents	**Prescribing recommendations**
Potent CYP3A4 inhibitors, e.g. Itraconazole Ketoconazole Posaconazole Voriconazole Erythromycin Clarithromycin	**Contraindicated with EZESIM**

Telithromycin HIV protease inhibitors (e.g. nelfinavir) Boceprevir Telaprevir Nefazodone Cobicistat Ciclosporin Danazol Gemfibrozil	
Other Fibrates Fusidic acid	Not recommended with EZESIM
Niacin (nicotinic acid) (≥ 1 g/day)	For Asian patients, not recommended with EZESIM
Amiodarone Amlodipine Verapamil Diltiazem Niacin (≥1 g/day) Elbasvir Grazoprevir	Do not exceed 10/20 mg EZESIM daily
Lomitapide	For patients with HoFH, do not exceed 10/40 mg EZESIM daily
Daptomycin	It should be considered to temporarily suspend EZESIM in patients taking daptomycin unless the benefits of concomitant administration outweigh the risk (see section 4.4)
Ticagrelor	Doses greater than 10/40 mg EZESIM daily are not recommended
Grapefruit juice	Avoid grapefruit juice when taking EZESIM

Effects of other medicinal products on EZESIM
EZESIM
Niacin: Concomitant EZESIM (10/20 mg daily for 7 days) caused a small increase in the mean AUCs of niacin and nicotinuric acid administered as NIASPAN extended-release tablets (1000 mg for 2 days and 2000 mg for 5 days following a low-fat breakfast). Concomitant NIASPAN slightly increased the mean AUCs of ezetimibe, total ezetimibe, simvastatin and simvastatin acid (see sections 4.2 and 4.4).
Drug interaction data with higher doses of simvastatin have not been investigated.

Ezetimibe
Antacids: Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

Cholestyramine: Concomitant cholestyramine administration decreased the mean area under the curve (AUC) of total ezetimibe (ezetimibe + ezetimibe glucuronide). The incremental LDL-C reduction due to adding EZESIM to cholestyramine may be lessened by this interaction (see section 4.2).

Ciclosporin: With a single 10-mg dose of ezetimibe, it resulted in an increase in the mean AUC for total ezetimibe. With daily administration of 20 mg ezetimibe for 8 days with a single 100-mg dose of ciclosporin on Day 7 resulted in increase in ciclosporin AUC. There is no data on the effect of co-administered ezetimibe on ciclosporin exposure in renal transplant patients. Concomitant administration of EZESIM with ciclosporin is contraindicated (see section 4.3).

Fibrates: Concomitant fenofibrate or gemfibrozil administration increased total ezetimibe concentrations. Although these increases are not considered clinically significant, co-administration of EZESIM with gemfibrozil is contraindicated and with other fibrates is not recommended (see sections 4.3 and 4.4).

Simvastatin
Simvastatin is a substrate of cytochrome P450 3A4. Potent inhibitors of cytochrome P450 3A4 increase the risk of myopathy and rhabdomyolysis by increasing the concentration of HMG-CoA reductase inhibitory activity in plasma during simvastatin therapy. Such inhibitors include itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, nefazodone, and medicinal products containing cobicistat. Concomitant administration of itraconazole resulted in a more than 10-fold increase in exposure to simvastatin acid (the active beta-hydroxyacid metabolite). Telithromycin caused an 11-fold increase in exposure to simvastatin acid.

Combination with itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin, nefazodone, and medicinal products containing cobicistat is contraindicated, as well as gemfibrozil, ciclosporin, and danazol (see section 4.3). If treatment with potent CYP3A4 inhibitors (agents that increase AUC approximately 5-fold or greater) is unavoidable, therapy with EZESIM must be suspended (and use of an alternative statin considered) during the course of treatment. Caution should be exercised when combining EZESIM with certain other less potent CYP3A4 inhibitors: fluconazole, verapamil, or diltiazem (see sections 4.2 and 4.4).

Other fibrates: The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of the individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDLC may be obtained. Combinations of fibrates with simvastatin have been used without myopathy in small short-term clinical studies with careful monitoring.

Amiodarone: In a clinical trial, myopathy was reported in 6% of patients receiving simvastatin 80 mg and amiodarone. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with amiodarone.

Ticagrelor: Co-administration of ticagrelor with simvastatin increased simvastatin Cmax and AUC and increased simvastatin acid Cmax and AUC with some individual increases equal to 2- to 3-fold. Co-administration of ticagrelor with doses of simvastatin exceeding 40 mg daily could cause adverse reactions of simvastatin and should be weighed against potential benefits. There was no effect of simvastatin on ticagrelor plasma levels. The concomitant use of ticagrelor with doses of simvastatin greater than 40 mg is not recommended.

Fluconazole: Rare cases of rhabdomyolysis associated with concomitant administration of simvastatin and fluconazole have been reported (see section 4.4).

Ciclosporin: The risk of myopathy/rhabdomyolysis is increased by concomitant administration of ciclosporin with EZESIM; therefore, use with ciclosporin is contraindicated (see sections 4.3 and 4.4). Although the mechanism is not fully understood, ciclosporin has been shown to increase the AUC of

HMG-CoA reductase inhibitors. The increase in AUC for simvastatin acid is presumably due, in part, to inhibition of CYP3A4 and/or OATP1B1.

Danazol: The risk of myopathy and rhabdomyolysis is increased by concomitant administration of danazol with EZESIM; therefore, use with danazol is contraindicated (see sections 4.3 and 4.4).

Gemfibrozil: Gemfibrozil increases the AUC of simvastatin acid by 1.9-fold, possibly due to inhibition of the glucuronidation pathway and/or OATP1B1 (see sections 4.3 and 4.4). Concomitant administration with gemfibrozil is contraindicated.

Fusidic acid: The risk of myopathy, including rhabdomyolysis, may be increased by the concomitant administration of systemic fusidic acid with statins. The mechanism of this interaction (whether it is pharmacodynamics or pharmacokinetic, or both) is yet unknown. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination. Co-administration of this combination may cause increased plasma concentrations of both agents.
If treatment with systemic fusidic acid is necessary, EZESIM treatment should be discontinued throughout the duration of the fusidic acid treatment. **Also see section 4.4.**

Amiodarone: The risk of myopathy and rhabdomyolysis is increased by concomitant administration of amiodarone with simvastatin (see section 4.4). Myopathy was reported in patients receiving simvastatin 80 mg and amiodarone. Therefore, the dose of EZESIM should not exceed 10/20 mg daily in patients receiving concomitant medication with amiodarone.

Calcium Channel Blockers
- Verapamil or diltiazem: In a clinical trial, patients on diltiazem treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with verapamil or diltiazem.
- Amlodipine: In a clinical trial, patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. The dose of simvastatin should not exceed 40 mg daily in patients receiving concomitant medication with amlodipine.
- Niacin (≥1g/day): The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) ≥ 1g/day. Cases of myopathy/rhabdomyolysis have been observed with simvastatin co-administered with lipid-modifying doses (≥ 1 g/day) of niacin.

Lomitapide: The risk of myopathy and rhabdomyolysis may be increased by concomitant administration of lomitapide with simvastatin (see sections 4.3 and 4.4). Therefore, in patients with HoFH, the dose of EZESIM must not exceed 10/40 mg daily in patients receiving concomitant medication with lomitapide.

Moderate Inhibitors of CYP3A4: Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 concomitantly with EZESIM, particularly higher EZESIM doses, may have an increased risk of myopathy (see section 4.4).

Inhibitors of the Transport Protein OATP1B1: Simvastatin acid is a substrate of the transport protein OATP1B1. Concomitant administration of medicinal products that are inhibitors of the transport protein OATP1B1 may lead to increased plasma concentrations of simvastatin acid and an increased risk of myopathy (see sections 4.3 and 4.4).

Inhibitors of Breast Cancer Resistant Protein (BCRP): Concomitant administration of medicinal products that are inhibitors of BCRP, including products containing elbasvir or grazoprevir, may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy (see sections 4.2 and 4.4).

Grapefruit juice: Grapefruit juice inhibits cytochrome P450 3A4. Concomitant intake of large quantities (over 1 litre daily) of grapefruit juice and simvastatin resulted in an increase in exposure to simvastatin acid. Intake of 240 mL of grapefruit juice in the morning and administration of simvastatin in the evening also resulted in an increase. Intake of grapefruit juice during treatment with EZESIM should therefore be avoided.

Colchicine: There have been reports of myopathy and rhabdomyolysis with the concomitant administration of colchicine and simvastatin, in patients with renal impairment. Close clinical monitoring of such patients taking this combination is advised.

Rifampicin: Because rifampicin is a potent CYP3A4 inducer, patients undertaking long-term rifampicin therapy (e.g. treatment of tuberculosis) may experience loss of efficacy of simvastatin. The area under the plasma concentration curve (AUC) for simvastatin acid was decreased with concomitant administration of rifampicin.

Daptomycin: The risk of myopathy and/or rhabdomyolysis may be increased by concomitant administration of HMG-CoA reductase inhibitors (e.g. simvastatin and ezetimibe/simvastatin) and daptomycin (see section 4.4).

Effects of EZESIM on the pharmacokinetics of other medicinal products
Ezetimibe
It has been shown that ezetimibe does not induce cytochrome P450 drug metabolising enzymes. No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolised by cytochromes P450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase.

Anticoagulants: Concomitant administration of ezetimibe (10 mg once daily) had no significant effect on bioavailability of warfarin and prothrombin time in a study of twelve healthy adult males. However, there have been post-marketing reports of increased International Normalised Ratio (INR) in patients who had ezetimibe added to warfarin or fluindione. If EZESIM is added to warfarin, another coumarin anticoagulant, or fluindione, INR should be appropriately monitored (see section 4.4).

Simvastatin: Simvastatin does not have an inhibitory effect on cytochrome P450 3A4. Therefore, simvastatin is not expected to affect plasma concentrations of substances metabolised via cytochrome P450 3A4.
Oral anticoagulants: Simvastatin 20-40 mg/day modestly potentiated the effect of coumarin anticoagulants: the prothrombin time, reported as International Normalized Ratio (INR), increased. Very rare cases of elevated INR have been reported. In patients taking coumarin anticoagulants, prothrombin time should be determined before starting EZESIM and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on coumarin anticoagulants. If the dose of EZESIM is changed or discontinued, the same procedure should be repeated. Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking anticoagulants.

<u>Paediatric population</u>
Interaction data have only been performed in adults.

4.6 Fertility, pregnancy and lactation
<u>Pregnancy</u>
Atherosclerosis is a chronic process, and ordinarily discontinuation of lipid-lowering drugs during pregnancy should have little impact on the long-term risk associated with primary hypercholesterolaemia.

EZESIM
EZESIM is contraindicated during pregnancy. No clinical data are available on the use of EZESIM during pregnancy. Animal data on combination therapy have demonstrated reproduction toxicity.

Simvastatin
The safety of simvastatin in pregnant women has not been established. No clinical data with simvastatin is available in pregnant women. Rare reports of congenital anomalies following intrauterine exposure to HMG-CoA reductase inhibitors have been received.
Maternal treatment with simvastatin may reduce the foetal levels of mevalonate which is a precursor of cholesterol biosynthesis. For this reason, EZESIM must not be used in women who are pregnant, trying to become pregnant or suspect they are pregnant. Treatment with EZESIM must be suspended for the duration of pregnancy or until it has been determined that the woman is not pregnant (see section 4.3).

Ezetimibe
No clinical data are available on the use of ezetimibe during pregnancy.

Breast-feeding
EZESIM is contraindicated during lactation. Data on rats have shown that ezetimibe is excreted into breast milk. It is not known if the active components of EZESIM are secreted into human breast milk (see section 4.3).

Fertility
Ezetimibe
No clinical data are available on the effects of ezetimibe on human fertility. Ezetimibe had no effect on the fertility of male or female rats.

Simvastatin
No clinical data are available on the effects of simvastatin on human fertility. Simvastatin had no effect on the fertility of rats male and female.

4.7 Effects on ability to drive and use machines
No data on the effects on the ability to drive and use machines is available. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been reported.

4.8 Undesirable effects
These reactions are presented in Table 1 by system organ class and by frequency.
The frequencies of adverse events are ranked according to the following: Very common, Common, Uncommon, Rare, Very Rare including isolated reports, and Not Known.

Table 1
Adverse Reactions

System organ class Frequency	**Adverse reaction**
Blood and lymphatic system disorders	
Not Known	thrombocytopaenia; anaemia
Immune system disorders	
Very Rare	anaphylaxis
Not Known	hypersensitivity
Metabolism and nutrition disorders	
Not Known	decreased appetite
Psychiatric disorders	

Uncommon	sleep disorder; insomnia
Not Known	depression
Nervous system disorders	
Uncommon	dizziness; headache; paraesthesia
Not Known	peripheral neuropathy; memory impairment
Eye disorders	
Rare	vision blurred; visual impairment
Vascular disorders	
Not Known	hot flush; hypertension
Respiratory, thoracic and mediastinal disorders	
Not Known	cough; dyspnoea; interstitial lung disease (see section 4.4)
Gastrointestinal disorders	
Uncommon	abdominal pain; abdominal discomfort; abdominal pain upper; dyspepsia; flatulence; nausea; vomiting; abdominal distension; diarrhoea; dry mouth; gastroesophageal reflux disease
Not Known	constipation; pancreatitis; gastritis
Hepatobiliary disorders	
Not Known	hepatitis/jaundice; fatal and non-fatal hepatic failure; cholelithiasis; cholecystitis
Skin and subcutaneous tissue disorders	
Uncommon	pruritus; rash; urticaria
Very Rare	lichenoid drug eruptions
Not Known	alopecia; erythema multiforme; angioedema
Musculoskeletal and connective tissue disorders	
Common	myalgia
Uncommon	arthralgia; muscle spasms; muscular weakness; musculoskeletal discomfort; neck pain; pain in extremity; back pain; musculoskeletal pain
Very Rare	muscle rupture
Not Known	muscle cramps; myopathy* (including myositis); rhabdomyolysis with or without acute renal failure (see section 4.4); tendinopathy, sometimes complicated by rupture; immune-mediated necrotising myopathy (IMNM)**
Reproductive system and breast disorders	
Very Rare	gynaecomastia
Not Known	erectile dysfunction
General disorders and administration site conditions	
Uncommon	asthenia; chest pain; fatigue; malaise; oedema peripheral
Not Known	pain
Investigations	
Common	ALT and/or AST increased; blood CK increased

Uncommon	blood bilirubin increased; blood uric acid increased; gamma-glutamyltransferase increased; international normalised ratio increased; protein urine present; weight decreased
Not Known	elevated alkaline phosphatase; liver function test abnormal

* Myopathy occurred commonly in patients treated with simvastatin 80 mg/day compared to patients treated with 20 mg/day (see sections 4.4 and 4.5).
** There have been very rare reports of immune-mediated necrotising myopathy (IMNM), an autoimmune myopathy, during or after treatment with some statins. IMNM is clinically characterised by: persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment; muscle biopsy showing necrotising myopathy without significant inflammation; improvement with immunosuppressive agents (see section 4.4).

Paediatric population
In adolescent (10 to 17 years of age) patients with heterozygous familial hypercholesterolaemia, elevations of ALT and/or AST (≥ 3X ULN, consecutive) were observed in the ezetimibe/simvastatin patients. No cases of myopathy were reported.

Patients with Chronic Kidney Disease
The safety profiles were comparable during a median follow-up period of 4.9 years. Consecutive elevations of transaminases occurred in patients treated with EZESIM (see section 4.4). There were no statistically significant increases in the incidence of pre-specified adverse events, including cancer, hepatitis, cholecystectomy or complications of gallstones or pancreatitis.

Laboratory Values
These elevations of serum transaminases (ALT and/or AST) were generally asymptomatic, not associated with cholestasis, and returned to baseline after discontinuation of therapy or with continued treatment (see section 4.4).
Clinically important elevations of CK were seen in patients treated with EZESIM.

Post-marketing Experience
An apparent hypersensitivity syndrome has been reported rarely which has included some of the following features: angioedema, lupus-like syndrome, polymyalgia rheumatica, dermatomyositis, vasculitis, thrombocytopaenia, eosinophilia, red blood cell sedimentation rate increased, arthritis and arthralgia, urticaria, photosensitivity reaction, pyrexia, flushing, dyspnoea and malaise.
Increases in HbA1c and fasting serum glucose levels have been reported with statins, including simvastatin.
There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use, including simvastatin. The reports are generally nonserious, and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).
The following additional adverse events have been reported with some statins:
• Sleep disturbances, including nightmares
• Sexual dysfunction
• Diabetes mellitus: Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI > 30 kg/m^2, raised triglycerides, history of hypertension).

4.9 Overdose
EZESIM

In the event of an overdose, symptomatic and supportive measures should be employed. Co-administration of ezetimibe (1000 mg/kg) and simvastatin (1000 mg/kg) was well-tolerated in acute, oral toxicity data in mice and rats. No clinical signs of toxicity were observed in these animals. The estimated oral LD_{50} for both species was ezetimibe ≥ 1000 mg/kg/simvastatin ≥ 1000 mg/kg.

<u>Ezetimibe</u>
Administration of ezetimibe, 50 mg/day to healthy patient for up to 14 days, or 40 mg/day to 18 patients with primary hypercholesterolaemia for up to 56 days, was generally well tolerated. A few cases of overdosage have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious. In animals, no toxicity was observed after single oral doses of 5000 mg/kg of ezetimibe in rats and mice and 3000 mg/kg in dogs.

<u>Simvastatin</u>
A few cases of overdosage have been reported; the maximum dose taken was 3.6 g. All patients recovered without sequelae.

5. Pharmacological properties

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: HMG-CoA reductase inhibitors in combination with other lipid modifying agents, ATC code: C10BA02
EZESIM (ezetimibe/simvastatin) is a lipid-lowering product that selectively inhibits the intestinal absorption of cholesterol and related plant sterols and inhibits the endogenous synthesis of cholesterol.

<u>Mechanism of action</u>
EZESIM
Plasma cholesterol is derived from intestinal absorption and endogenous synthesis. EZESIM contains ezetimibe and simvastatin, two lipid-lowering compounds with complementary mechanisms of action. EZESIM reduces elevated total cholesterol (total-C), LDL-C, apolipoprotein B (Apo B), triglycerides (TG), and non-high-density lipoprotein cholesterol (non-HDL-C), and increases high-density lipoprotein cholesterol (HDL-C) through dual inhibition of cholesterol absorption and synthesis.

Ezetimibe
Ezetimibe inhibits the intestinal absorption of cholesterol. Ezetimibe is orally active and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g. statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol transporter, Niemann-Pick C1-Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols.
Ezetimibe localises at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver; statins reduce cholesterol synthesis in the liver and together these distinct mechanisms provide complementary cholesterol reduction. Ezetimibe inhibited intestinal cholesterol absorption.
Ezetimibe inhibited the absorption of [^{14}C]-cholesterol with no effect on the absorption of triglycerides, fatty acids, bile acids, progesterone, ethinyl estradiol, or fat soluble vitamins A and D.

Simvastatin
After oral ingestion, simvastatin, which is an inactive lactone, is hydrolysed in the liver to the corresponding active β-hydroxyacid form which has a potent activity in inhibiting HMG-CoA reductase (3 hydroxy - 3 methylglutaryl CoA reductase). This enzyme catalyses the conversion of HMG-CoA to mevalonate, an early and rate-limiting step in the biosynthesis of cholesterol.
Simvastatin has been shown to reduce both normal and elevated LDL-C concentrations. LDL is formed from very-low-density protein (VLDL) and is catabolized predominantly by the high affinity LDL receptor. The mechanism of the LDL-lowering effect of simvastatin may involve both reduction of VLDL-cholesterol (VLDL-C) concentration and induction of the LDL receptor, leading to reduced

production and increased catabolism of LDL-C. Apolipoprotein B also falls substantially during treatment with simvastatin. In addition, simvastatin moderately increases HDL-C and reduces plasma TG. As a result of these changes, the ratios of total- to HDL-C and LDL- to HDL-C are reduced.

5.2 Pharmacokinetic properties
No clinically significant pharmacokinetic interaction was seen when ezetimibe was co-administered with simvastatin.

Absorption
EZESIM
EZESIM is bioequivalent to co-administered ezetimibe and simvastatin.

Ezetimibe
After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe-glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection.
Concomitant food administration (high-fat or non-fat meals) had no effect on the oral bioavailability of ezetimibe when administered as 10-mg tablets.

Simvastatin
The availability of the active β-hydroxyacid to the systemic circulation following an oral dose of simvastatin was found to be less than 5% of the dose, consistent with extensive hepatic first-pass extraction. The major metabolites of simvastatin present in human plasma are the β-hydroxyacid and four additional active metabolites.
Relative to the fasting state, the plasma profiles of both active and total inhibitors were not affected when simvastatin was administered immediately before a test meal.

Distribution
Ezetimibe
Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Simvastatin
Both simvastatin and the β-hydroxyacid are bound to human plasma proteins (95%).
The pharmacokinetics of single and multiple doses of simvastatin showed that no accumulation of drug occurred after multiple dosing. In all of the above pharmacokinetic data, the maximum plasma concentration of inhibitors occurred 1.3 to 2.4 hours post-dose.

Biotransformation
Ezetimibe
Ezetimibe is metabolised primarily in the small intestine and liver via glucuronide conjugation (a phase II reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase I reaction) has been observed in all species evaluated. Ezetimibe and ezetimibe-glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20% and 80 to 90% of the total drug in plasma, respectively. Both ezetimibe and ezetimibe-glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Simvastatin

Simvastatin is an inactive lactone which is readily hydrolysed to the corresponding β-hydroxyacid, a potent inhibitor of HMG-CoA reductase. Hydrolysis takes place mainly in the liver; the rate of hydrolysis in human plasma is very slow.
In man simvastatin is well absorbed and undergoes extensive hepatic first-pass extraction. The extraction in the liver is dependent on the hepatic blood flow. The liver is its primary site of action, with subsequent excretion of drug equivalents in the bile. Consequently, availability of active drug to the systemic circulation is low.
Following an intravenous injection of the β-hydroxyacid metabolite, its half-life averaged 1.9 hours.

Elimination
Ezetimibe
Following oral administration of ^{14}C-ezetimibe (20 mg), total ezetimibe accounted for approximately 93% of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the faeces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Simvastatin
Simvastatin acid is taken up actively into the hepatocytes by the transporter OATP1B1.
Simvastatin is a substrate of the efflux transporter BCRP.
Following an oral dose of radioactive simvastatin, 13% of the radioactivity was excreted in the urine and 60% in the faeces within 96 hours. The amount recovered in the faeces represents absorbed drug equivalents excreted in bile as well as unabsorbed drug. Following an intravenous injection of the β-hydroxyacid metabolite, an average of only 0.3% of the IV dose was excreted in urine as inhibitors.

Special Populations
Paediatric Population
The absorption and metabolism of ezetimibe are similar between children and adolescents (10 to 18 years) and adults. Based on total ezetimibe, there are no pharmacokinetic differences between adolescents and adults. Pharmacokinetic data in the paediatric population < 10 years of age are not available. Clinical experience in paediatric and adolescent patients includes patients with HoFH, HeFH, or sitosterolaemia (see section 4.2).

Elderly
Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥ 65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and younger patient treated with ezetimibe (see section 4.2).

Hepatic impairment
After a single 10-mg dose of ezetimibe, the mean AUC for total ezetimibe was increased in patients with mild hepatic impairment (Child-Pugh score 5 or 6). The mean AUC for total ezetimibe was increased on Day 1 and Day 14. No dosage adjustment is necessary for patients with mild hepatic impairment. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child-Pugh score > 9) hepatic impairment, ezetimibe is not recommended in these patients (see sections 4.2 and 4.4).

Renal impairment
Ezetimibe
After a single 10-mg dose of ezetimibe in patients with severe renal disease, the mean AUC for total ezetimibe was increased, (see section 4.2).

Simvastatin
In patients with severe renal impairment (creatinine clearance < 30 mL/min), the plasma concentrations of total inhibitors after a single dose of a related HMG-CoA reductase inhibitor were higher
Gender

Plasma concentrations for total ezetimibe are slightly higher in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe.

SLCO1B1 polymorphism
Carriers of the SLCO1B1 gene c.521T > C allele have lower OATP1B1 activity. The mean exposure (AUC) of the main active metabolite, simvastatin acid is 120% in heterozygote carriers (CT) of the C allele and 221% in homozygote (CC) carriers relative to that of patients who have the most common genotype (TT). The C allele has a frequency of 18% in the European population. In patients with SLCO1B1 polymorphism there is a risk of increased exposure of simvastatin acid, which may lead to an increased risk of rhabdomyolysis (see section 4.4).

6. Pharmaceutical particulars
6.1 Incompatibilities
Not applicable.

6.2 Shelf life
2 years.

6.3 Special precautions for storage
Store below 30°C. Protect from light.

6.4 Nature and contents of container
Alu/OPA/PE Blister of 10's Tablets. Such three blisters are packed into an outer carton with pack insert.

6.5 Special precautions for disposal and other handling
No special requirements.

7. Name And Address Of Manufacturer

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10. Date Of Revision Of Text

March 2025

font size min size: 7, as measure in Times New Roman